CLEAN AIR FOR BLUE SKIES: CASE STUDY

TEAM MULTISIM

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Problem Statement

Sustainable Air Quality Management for Rourkela

Rourkela, as one of Odisha's largest industrial and educational hubs, faces significant air quality challenges.

CITY'S OVERVIEW



What is Rourkela

A major urban-industrial center developed around an integrated steel complex, with township and plant zones shaping the city's layout and identity.

Key Statistics

Population: ~620,000 (2025)
Area: 170 km²
Literacy rate: 83%
Urban/rural: 70% urban, 30% rural

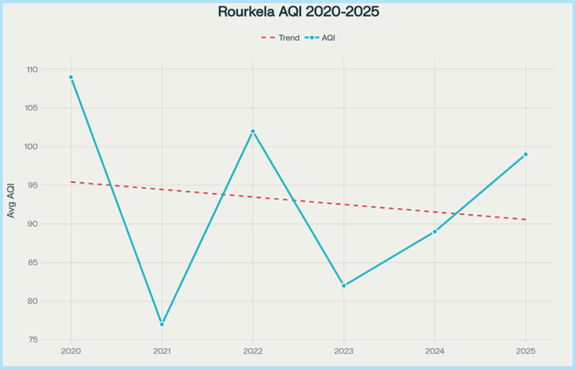
Industries Present

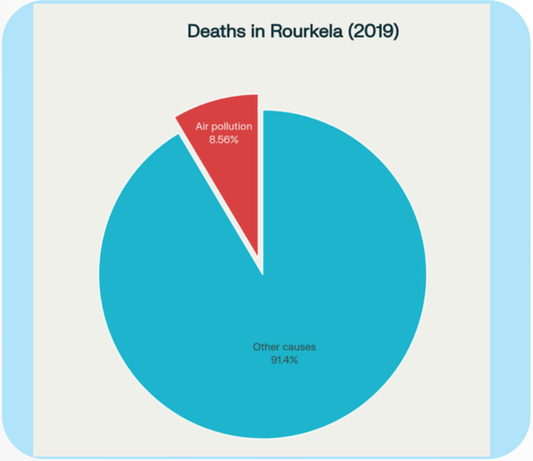
SAIL-run Rourkela Steel Plant ,refractories, cement, fertilizers and engineering units

Environmental Profile

Annual AQI: ~105 (Moderate–Poor)
Main pollutants: $PM_{2.5}$, NO_2, SO_2
Green cover: ~22% of city area

AQI TRENDS OF PAST 5 YEARS





AIR QUALITY ANALYSIS OF STEEL CITIES OF INDIA

Problem Area	Rourkela	Jamshedpur	Bhilai	Durgapur	Visakhapatnam
1. High AQI Days	Medium — usually Moderate AQI (50–80) in monsoon; winter can spike to Poor/Unhealthy (episodes >200 reported)	Medium — generally Moderate/Good; PM2.5 often 16–24 µg/m³; occasional Moderate AQI 50–80	Medium — typically Moderate (AQI ~60–90) with regional dispersion effects	Medium — Moderate most days; winter dispersion degrades AQI	Medium — coastal dispersion helps but episodes reach Poor; PM10 often elevated near sources
2. PM2.5 Baseline (recent)	Low — ~16–18 µg/m³; near but above WHO 15 guideline	Low — ~16–24 µg/m³ on many days	Medium — mid-20s to low-30s µg/m³ typical for industrial belts	Medium — mid-20s to high-20s µg/m³ regularly	Medium — upper-20s to 30s µg/m³; sea breeze moderates but not near sources
3. PM10 Levels	Medium — ~40–55 µg/m³ recently; road dust/handling contribute	Low — ~27–38 µg/m³; often below 24-hr NAAQS	Medium — ~40–60 µg/m³ in heavy-industry corridors	Medium — ~50–70 µg/m³ around industrial zones	High — >80–110 µg/m³ near port/industrial belts during episodes
4. Episodic Spikes/Seasonality	High — winter stagnation yields sharp spikes; CPCB flagged Poor/Unhealthy events in 2025	Medium — winter inversions raise PM; spikes shorter/milder than peers	Medium — winter haze elevates AQI into upper-Moderate/Poor	Medium — cold-season episodes push toward Poor; monsoon relief evident	Medium — sea breeze aids dispersion yet thermal/port plumes cause Poor days
5. Compliance Risk vs Standards	Medium — daily PM2.5 often near/over WHO; generally within Indian NAAQS recently	Low — frequent daily NAAQS compliance; borderline to above WHO at times	Medium — often meets daily NAAQS but winter exceedance risk	Medium — mixed compliance; PM10 exceedance risk in industrial corridors	Medium/High — elevated PM10 exceedance risk; PM2.5 near/over WHO on many days

HIGH — Indicates unfavorable conditions or higher risk

MEDIUM — Indicates moderate conditions or medium risk

LOW — Indicates relatively favorable conditions or lower risk

RECENT ROURKELA AIR QUALITY HIGHLIGHTS

1. In 2023, the CPCB designated Rourkela a "non-attainment" city for consistently exceeding national air quality standards due to steel, power, mining, and transport emissions.
2. 2025 Year-to-Date: Rourkela's average AQI of 99 represents a 9.8% deterioration compared to 2024, with 54% of days falling into the Moderate category and 45% into Poor or worse.
3. Peak AQI in winter months often exceeds 150 (Poor–Unhealthy), driven by industrial heating, lower dispersion, and dust re-suspension.

Rourkela's Environment at a Glance

Air Pollution

- **PM10** concentrations: 32.98-133.46 µg/m³ (above NAAQS standard of 60 µg/m³)
- **PM2.5** levels: Average 32.57 µg/m³ with peaks at 50.05 µg/m³
- **Rourkela Township** identified as **most polluted area**

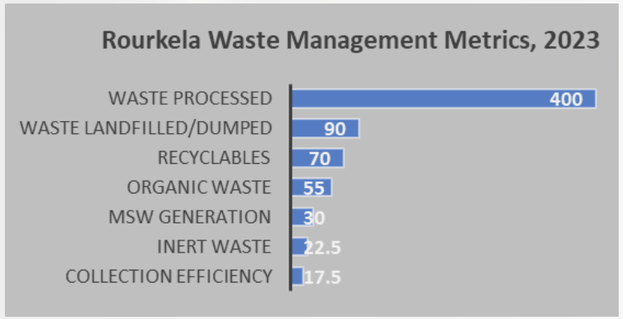
Water Pollution

- Brahmani River: **BOD levels increased to 6.6 mg/l downstream**, dissolved oxygen decreased significantly
- RSP **generates 84,000 m³/day wastewater**, with **fluoride exceeding limits by 130-850% and cyanide by 1,800%**

Heavy Metals: **Iron content 0.8-3.7 mg/l** (above 0.3 mg/l limit)

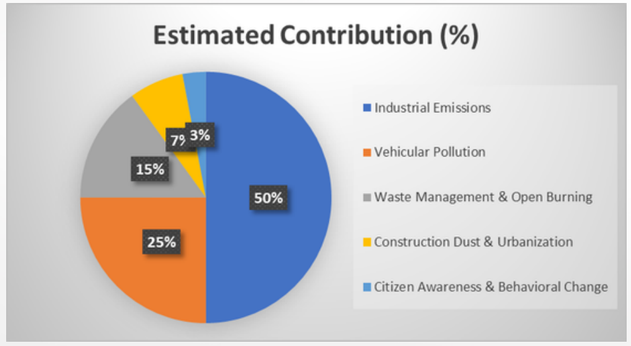
Health Impacts

- **Higher incidences of respiratory diseases**, tuberculosis, and gastric disorders in industrial areas
- **2.45 times increased odds of chronic respiratory conditions** in high pollution zones



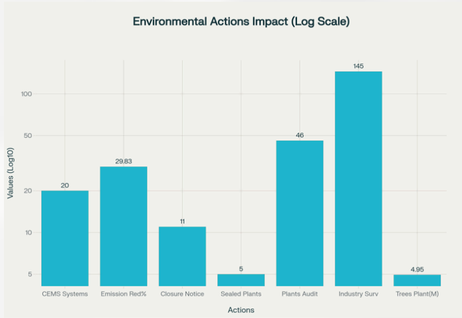
Rourkela's 2023 waste profile shows the bulk of activity in waste processed at 400 units, far outpacing landfilled or dumped waste at 90, which signals a strong tilt toward treatment over disposal. Recyclables and organic waste account for 70 and 55 respectively, indicating meaningful material recovery and composting potential within the stream. MSW generation is charted at 30 with inert waste at 22.5, while collection efficiency sits at 17.5, highlighting a key operational gap where improving collection could unlock higher processing and diversion outcomes

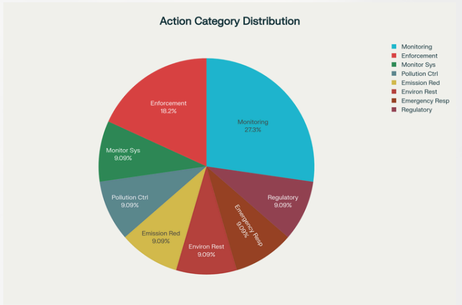
Rourkela's air pollution is led by industrial emissions at 50%, with vehicles adding 25%, together forming three-fourths of the burden. Waste and open burning account for 15%, while construction and urbanization add 7%, showing notable area-source effects in this industrial hub. Citizen awareness and behavioral change represent 3%, pointing to a smaller but important avenue alongside cleaner industry, transport controls, and better waste practices

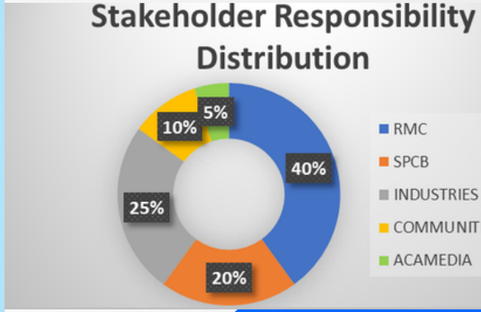


Major Actions Taken

Rourkela Steel Plant (SAIL-RSP)	- Installed Electro Static Precipitators, Bag Houses, Multi-Cyclones for dust/ particulate control - **20 Continuous Emission Monitoring Systems** with remote calibration - **Stack emission loads reduced by 29.83% below prescribed norms**
Pollution Control Board (OSPCB)	- Issued **closure notices to 11 polluting industrial units** (2024–25) - Sealed 5 sponge iron plants for non-compliance - **46 sponge iron plants under comprehensive environmental audit** - **145 industries mandated to install internet-based surveillance & CEMS** - **Air Quality Monitoring Stations** linked to CPCB/SPCB - Implemented **four-tier emergency response plan for high pollution** episodes
Other Industries	- **Sponge iron sector under closure orders, audits, and surveillance mandates**
Individuals Community Initiatives	- **Tree plantation (49.5 lakh trees planted)** enhancing local air quality and carbon sequestration







STAKEHOLDERS AND THEIR ROLES



Rourkela Municipal Corporation (RMC)

- **Manages urban planning, waste management, and civic infrastructure** to curb open burning and construction dust.
- **Implements traffic regulations and public awareness programs** to reduce vehicular emission

ODISHA
State Pollution Control Board, Odisha
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State Pollution Control Board (SPCB)

- **Monitors air quality** (PM2.5, PM10, NOx, SO2), enforces pollution control regulations and conducts inspection.
- **Regulates industrial emissions and coordinates action plans** to address AQI exceedances.



Industries (e.g., Rourkela Steel Plant)

- **Reduces emissions through pollution control technologies** (e.g., scrubbers)
- **Implements self-monitoring and offsets impacts** via green initiatives (e.g., tree planting).



Local Communities

- Participates in **waste segregation and reports pollution** issues to authorities
- **Adopts behavioral changes** (e.g., reduced vehicle use, clean practices).



Academia (e.g., NIT Rourkela)

- **Conduct research on air quality trends and innovative solutions.**
- **Provides training and policy recommendations for sustainable management.**

SOLUTIONS

Industrial Solutions

Upgrade Pollution Control Devices: Mandate advanced filters, scrubbers, baghouses, and continuous emission monitoring in all large industries.

Adopt Cleaner Fuels: Promote partial switching from coal to piped natural gas (PNG), biomass, or green hydrogen for process heat.

Encourage Best Available Techniques (BAT): Incentivize industry-wide adoption of BAT for emission reduction and energy efficiency.

Municipal and Urban Management

- Expand Green Buffers: Fast-track tree plantation efforts, prioritizing air pollution hotspots and roadside corridors.
- Public Transit & Mobility: Accelerate adoption of electric/CNG buses, build cycling paths, and discourage private vehicle use with low emission zones.
- Dust & Construction Controls: Enforce green mesh netting, regular sprinkling, and strict debris transport regulations at construction and demolition sites.
- Waste Management: Set up decentralized composting and dedicated facilities for construction/demolition waste, discouraging open burning.

Technological Innovations

- Real-Time AQI Apps: Partner with local tech/academic institutions to develop a city-specific air quality dashboard for public use.
- IoT and Smart City Tools: Use remote sensing, AI, and IoT for predictive pollution mapping and to inform traffic or emergency action plans.

Action Plan Phases

Short-Term Action Plan (0-3 Years): Foundation & Enforcement

Industrial Emissions

- install Continuous Emission Monitoring Systems (CEMS)
- Start a large-scale tree-planting initiative around industrial clusters

Vehicular Emissions

- zero-tolerance policy against polluting vehicles
- Prepare a Detailed Project Report (DPR) to modernize public transport

Construction Dust

- strictly enforce rules like covering sites with green netting, regular water sprinkling, and covering debris-carrying trucks.

Moderate-Term Action Plan (3-7 Years): Infrastructure & Transition

Industrial Emissions

Implement a phased mandate for the most polluting industries to upgrade their systems to the Best Available Techniques (BAT), including advanced filters and effluent treatment plants.

Vehicular Emissions

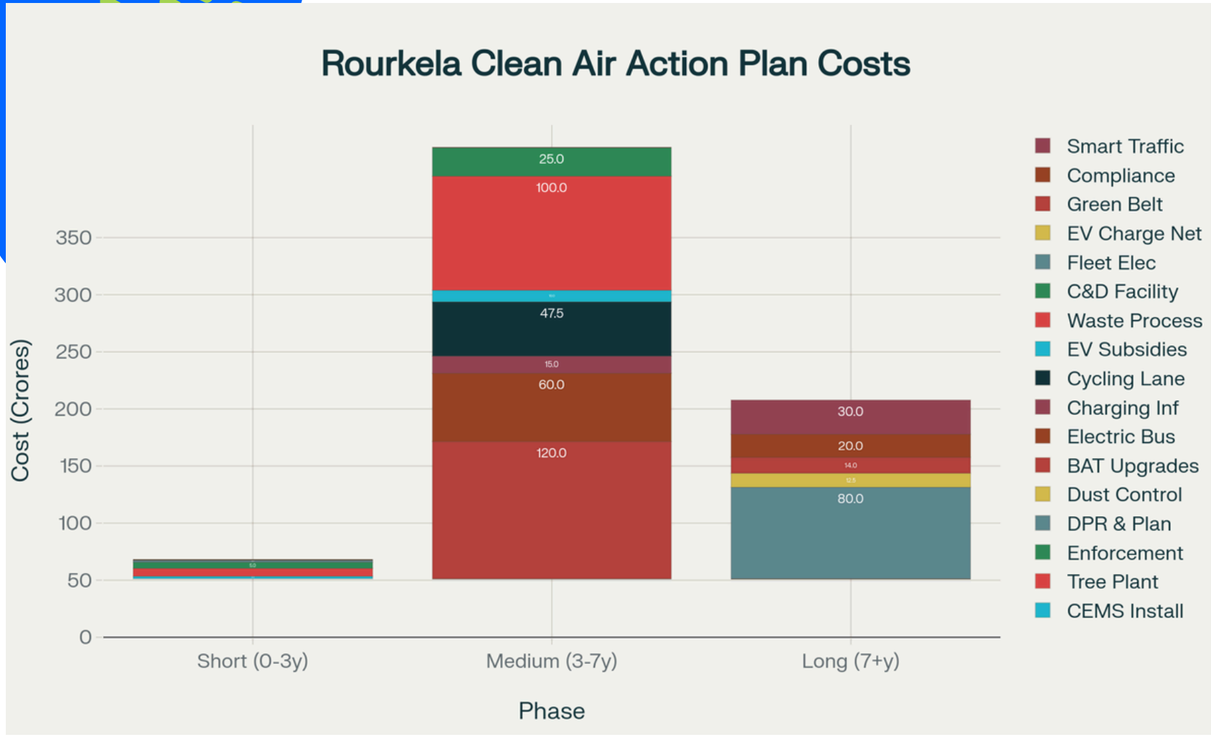
- construction of dedicated cycling lanes and safe footpaths to connect major parts of the city.
- Subsidise local businesses for EV

Waste Management

- dedicated facility to process and recycle waste from construction and demolition activities.

Long-Term Action Plan (7+ Years): Consolidation & Scale-up

- Transition the majority of the city's public transportation fleet to electric and other clean energy sources.
- reliable network of EV charging stations
- Dense green belt around industrial areas
- Ensure all industries are compliant with the highest pollution control standards.



Sector-wise Investment Distribution

- **Transport & Mobility (44.5%): ₹245.0 Crores**
- **Industrial Emissions (25.8%): ₹142.2 Crores**
- **Green Infrastructure (3.8%): ₹21.0 Crores**
- **Enforcement & Others (3.2%): ₹17.4 Crores**

Implementation Recommendations

- Phase funding in 3-year cycles aligned with central scheme allocations and performance reviews
- Prioritize PPP for waste management (₹125 Crores) to reduce upfront capital burden on ULB
- Leverage RSP's existing infrastructure for monitoring system integration and cost optimization
- Bundle transport projects with FAME-II and AMRUT funding for enhanced viability

Phase-wise Cost Breakdown

Short-term (0-3 years): ₹16.6 Crores

- **CEMS Installation (15 new stacks): ₹2.25 Crores** - Industrial emission monitoring systems for non-RSP industries, as RSP already has 20 CEMS units
- **Tree Planting (50 hectares): ₹7.0 Crores** - Buffer zones around industrial clusters at ₹130-150 per sq ft
- **Enforcement Infrastructure: ₹5.0 Crores** - Monitoring systems, checkpoints, and zero-tolerance policy implementation
- **DPR & Planning: ₹2.0 Crores** - Detailed project reports for transport modernization
- **Construction Dust Control (20 sites): ₹0.4 Crores** - Netting and sprinkling equipment

Medium-term (3-7 years): ₹377.5 Crores

- **BAT Upgrades (8 major industries): ₹120.0 Crores** - Best Available Techniques implementation beyond RSP
- **Electric Buses (30 units): ₹60.0 Crores** - E-bus procurement at ₹2 Crores per unit under FAME-II guidelines
- **Bus Charging Infrastructure (3 depots): ₹15.0 Crores** - Depot charging facilities
- **Cycling Lanes (25 km): ₹47.5 Crores** - Dedicated cycling infrastructure at ₹1.9 Crores per km
- **Waste Processing Facility (200 TPD): ₹100.0 Crores** - Municipal solid waste processing via PPP model
- **C&D Waste Facility: ₹25.0 Crores** - Construction and demolition waste recycling
- **EV Subsidies: ₹10.0 Crores** - Local business electric vehicle adoption incentives

Long-term (7+ years): ₹156.5 Crores

- **Complete Fleet Electrification (40 buses): ₹80.0 Crores** - Remaining public transport electrification
- **Smart Traffic System: ₹30.0 Crores** - Citywide intelligent transportation system
- **Compliance Monitoring: ₹20.0 Crores** - Advanced industrial compliance tracking systems
- **Green Belt Expansion (100 hectares): ₹14.0 Crores** - Dense green cover completion
- **EV Charging Network (50 stations): ₹12.5 Crores** - Public charging infrastructure at ₹25 lakh per station

THANK YOU

